

How DG Degrades the CTI, and How the DSDR Mitigates It

A complete guide with the results of the study

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1. The quantity: coordination time interval (CTI)

For fuse saving the recloser must open on its fast curve *before* the fuse starts to melt:

$$CTI = t_{MMT}(I_{fuse}) - t_{fast}(I_{rec})$$

- t_{MMT} : minimum melting time of the fuse at the current through the fuse.
- t_{fast} : fast operating time of the recloser at the current through the recloser.
- $CTI > 0$: the recloser wins, the fuse is saved. $CTI < 0$: the fuse melts first, fuse saving is lost.

Without DG the fuse and the recloser are in series and carry the *same* current, so the CTI is fixed by the shape of the two curves. With DG this is no longer true — that is the whole problem.

2. How the study measured it

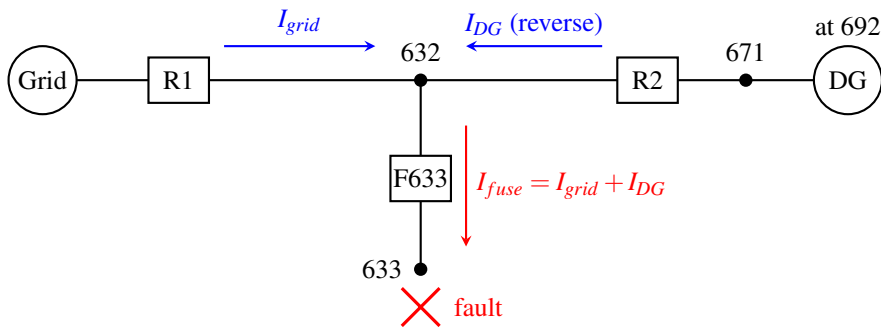
- Seven PowerFactory models, DG at node 692 at 0, 10, 25, 37, 50, 75 and 100 % of 4.05 MVA. The main model was not changed.
- Two faults, as in the reference method: **internal fault at 633** (bolted LLL; fuse F633 against R1) — the DG is *beyond* the recloser, between the fault and R2; **external fault at 684** (fuse F671-2 against R2) — the DG is on the same side as the fault, downstream of R2.
- Two protection schemes: **conventional** (single-setting R2, fuses of the design without DG, F633 250E) and **DSDR** (R2 with forward and reverse groups designed for 100 % DG, revised fuses, F633 400E).
- All times are PowerFactory's own relay and fuse times.

3. How DG degrades the CTI

3.1 The mechanism, step by step

The internal fault at 633 is used; the numbers are PowerFactory's, DG at 0 % and 100 %.

Step 1 — Where the currents flow. Picture the feeder as two sources feeding one fault:



By Kirchhoff's current law at node 632, everything that flows to the fault must pass F633. R1 carries only what the grid sends; R2 (in reverse) carries only what the DG sends. So the fuse carries the *sum*, each recloser only its *own part*. At 100 %: $I_{R1} = 3970$ A, $I_{R2} = 1500$ A (reverse), $I_{F633} = 5426$ A. (The sum is not exactly

3970 + 1500 because the two currents are phasors with slightly different angles and the loads also take current.) Without DG the fuse and R1 carry almost the same current (4102 and 4162 A; R1 also feeds the loads of the rest of the feeder).

Step 2 — Why the grid share even falls. The grid pushes current through its own impedance Z_g (source, substation transformer, regulator and the line to 632):

$$I_{grid} = \frac{E_{grid} - V_{632}}{Z_g}.$$

During the fault the voltage at 632 collapses. The DG injects current into the same node and holds V_{632} a little higher, so the voltage across Z_g is smaller and the grid sends less: I_{R1} falls from 4162 to 3970 A (−4.6 %). In addition the DG supplies part of the load before the fault (R1's load current falls from 588 to 285 A), which also lowers R1's current.

Step 3 — How each device's time reacts. Both curves are inverse (more current, less time), but the two devices see very different current changes:

Device	Current 0 %	Current 100 %	Change	Time 0 % → 100 %	Time change
F633 (melting)	4102 A	5426 A	+32 %	0.1018 → 0.0548 s	−46 %
R1 (fast)	4162 A	3970 A	−4.6 %	0.0982 → 0.1055 s	+7 %

Two reasons combine:

1. **The fuse current changes far more.** +32 % for the fuse against −5 % for R1 (Step 1 and 2).
2. **The fuse curve is steeper.** On a log–log plot the slope tells how many per cent the time changes for one per cent of current. Here the fuse's local slope is about −2.2 (melting is a heating process, close to $I^2t = \text{constant}$, i.e. slope −2), while R1's extremely inverse curve at about 5.5 times pickup is flatter, about −1.5, because it is approaching its definite-time end. (This is also why the method's fuse equation with $a = -1.8$ predicts $1.32^{-1.8} = 0.61$, a little less change than PowerFactory's 0.54: at these currents the real curve is steeper.)

Step 4 — Add it up. The change of the CTI is the change of the fuse time minus the change of the recloser time:

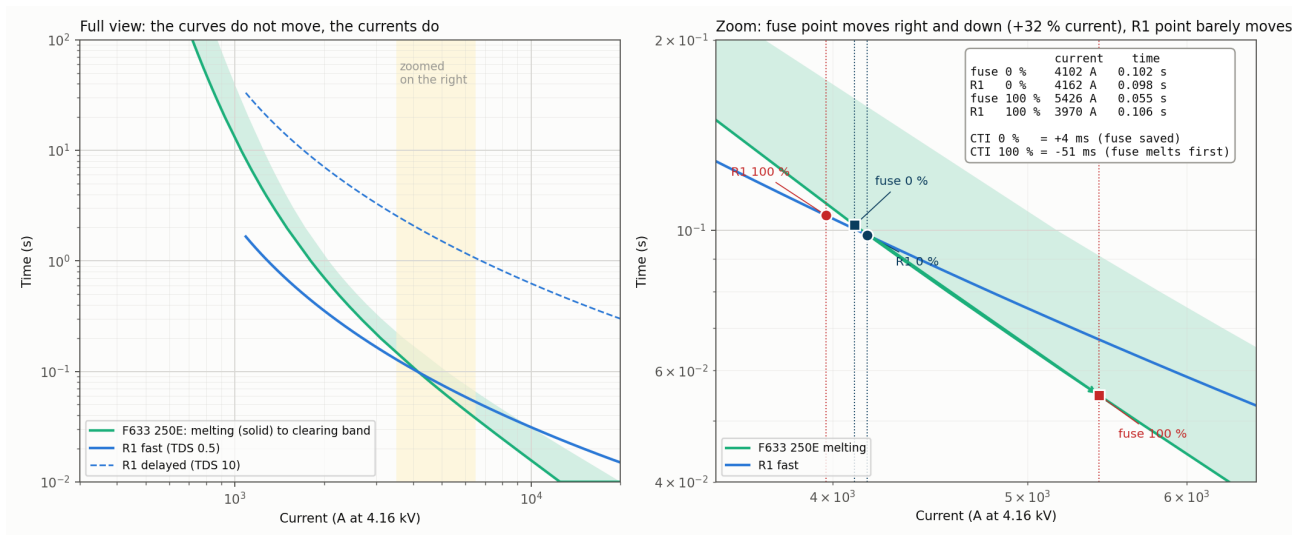
$$\Delta\text{CTI} = \Delta t_{MMT} - \Delta t_{fast} = (-47.0\text{ms}) - (+7.3\text{ms}) = -54.3\text{ms}.$$

About 87 % of the loss comes from the fuse becoming faster and 13 % from the recloser becoming slower. The CTI goes from +3.6 to −50.7 ms.

Step 5 — Why it turns negative so early at 633. The design without DG left only +4 ms. Already 10 % DG (0.41 MVA) adds 154 A to the fuse current; the melting time drops by 8 ms and the CTI becomes −5 ms. The zero crossing is at about 4 % penetration. A design with a larger starting margin survives longer — which is what the fuse revision does (Section 4).

Step 6 — The same at 684. For the fault at 684 the DG at 692 feeds through F671-2 but not through R2: the fuse current rises by 54 % (2574 → 3963 A) and its melting time falls by 69 % (0.470 → 0.148 s), while R2's current falls by 6 % and its fast time rises only from 0.067 to 0.072 s. The CTI falls from +403 to +75 ms — the same mechanism, but the large starting margin keeps it positive.

The same on the time–current characteristic (TCC).



TCC of the fault at 633 with the conventional protection: fuse F633 (250E) and R1. Left: full view. Right: zoom on the operating points at 0 % (dark) and 100 % DG (red).

How to read it:

- The **curves never move** — the settings and fuse are the same at every DG level. Only the **operating points** move, because the currents change.
- Each device is read at *its own* current: the fuse (square) on the green melting curve at the fuse current, R1 (circle) on the blue fast curve at R1's current. The vertical dotted lines mark these currents.
- **0 % DG:** the two currents are almost equal (4102 and 4162 A), the points sit almost on top of each other where the curves cross; the fuse square is just *above* the R1 circle: CTI = +4 ms, R1 trips first.
- **100 % DG:** the fuse square slides *down the green curve to the right* (more current, 5426 A, melts at 0.055 s), while the R1 circle moves only slightly *left and up* (less current, 3970 A, 0.106 s). The fuse is now far *below* R1: CTI = -51 ms, the fuse melts first.
- The vertical gap between square and circle *is* the CTI. DG pulls the two points apart sideways (different currents), and on falling curves sideways separation turns into a time gap of the wrong sign.

In short: DG makes the fuse and the recloser carry *different* currents. The fuse, nearer the fault, gets more; the recloser gets the same or less. The fuse therefore speeds up while the recloser does not, and the gap between them — the CTI — closes.

3.2 Internal fault at 633, conventional protection

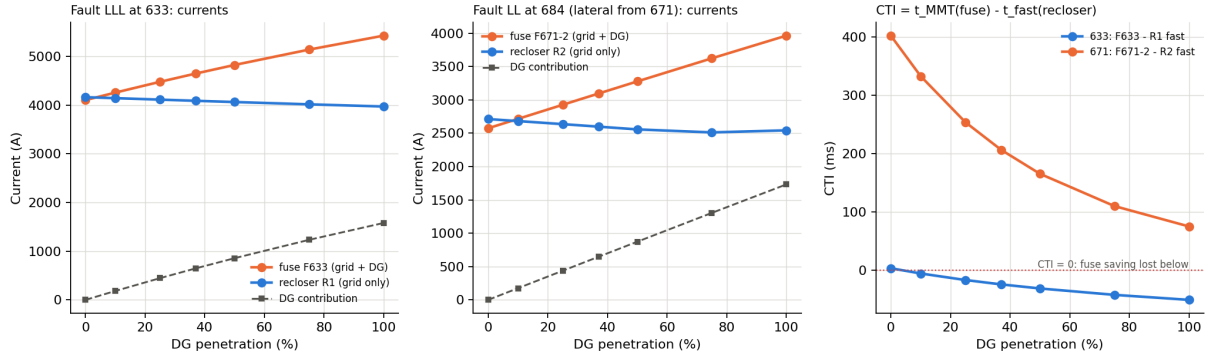
DG	DG (MVA)	I_{F633} (A)	I_{R1} (A)	t_{MMT} F633 (s)	t_{fast} R1 (s)	CTI (ms)
0 %	0.00	4102	4162	0.102	0.098	+4
10 %	0.41	4256	4142	0.094	0.099	-5
25 %	1.01	4478	4111	0.083	0.100	-17
37 %	1.50	4648	4088	0.077	0.101	-24
50 %	2.02	4824	4062	0.071	0.102	-31
75 %	3.04	5140	4015	0.062	0.104	-42
100 %	4.05	5426	3970	0.055	0.106	-51

The margin without DG was only +4 ms, so the CTI turns negative at about 4 % penetration.

3.3 External fault at 684, conventional protection

DG	I_{F671-2} (A)	I_{R2} (A)	t_{MMT} F671-2 (s)	t_{fast} R2 (s)	CTI (ms)
0 %	2574	2711	0.470	0.067	+403
25 %	2926	2636	0.323	0.069	+254
50 %	3276	2557	0.237	0.072	+166
75 %	3623	2511	0.184	0.073	+110
100 %	3963	2543	0.148	0.072	+75

Same mechanism (the DG at 692 feeds the fault at 684 through F671-2 but not through R2). The CTI falls by 81 % but stays positive, because the margin without DG was large.



Fuse current (grid + DG), recloser current (grid only), DG current and resulting CTI against DG penetration.

3.4 The other effects of rising penetration

- **Fault level rises**, most near the DG: 671/692 +64 %, 675 +59 %, 684 +55 %, 633 +32 %, 632 +38 %. Least behind the transformer (634, +12 %) and on the single-phase laterals (+19–20 %).
- **Reverse current through R2** for faults upstream of it grows: fault at 633, 66 A at 0 % to 1500 A at 100 %. A conventional (non-directional) R2 starts to trip on it from about 37 % — a trip for a fault outside its zone, which also disconnects the healthy part beyond 671 together with the DG.
- **Load flow reverses**. The power through R2 falls from 2460 kW towards 671 (0 %) to 755 kW towards 632 (100 %); the current through R1 falls from 588 to 285 A.
- **Loss of sensitivity**. The weakest fault in R2's zone (LG through 3 Ω) falls from 879 A to 368 A, below R2's 600 A start from about 75 %: R2 no longer sees it.

In one sentence: The fuse carries grid plus DG current while the recloser carries only the grid share; the fuse current therefore rises much more than the recloser current changes, the fuse curve is also steeper, so the fuse becomes faster far more quickly than the recloser and the CTI falls with every step of DG penetration.

4. How the DSDR mitigates it

4.1 Two actions that work together

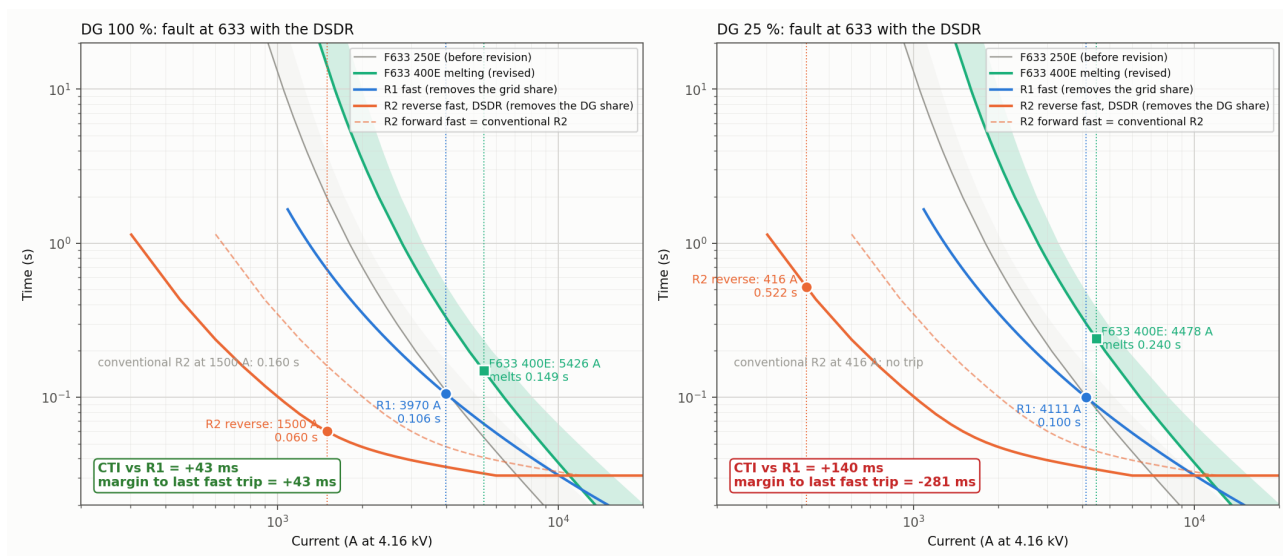
1. **A reverse setting group on R2**. For a fault at 633 the DG current flows backwards through R2. A conventional R2 is set for the large forward grid current and is slow or blind to this smaller current. The reverse group has a lower pickup (150/300 A plugs, curve from 300 A) and trips fast on it: at 100 % DG R2 reverse fast = 0.060 s (conventional R2: 0.160 s). R1 removes the grid share, R2 removes the DG share, so *both* sources are cut before the fuse melts.

2. **The fuse revision.** The fuse is made one or two sizes larger (F633 250E → 400E), so its melting curve moves up and the margin over R1 is restored. This is allowed only because the dial ranges were exhausted (step 9 of the method). The revision alone is not enough: with revised fuses but a conventional R2 only 31 of 39 cells are held; with the DSDR, 38.

4.2 Internal fault at 633: conventional against DSDR

DG	CTI against R1 (ms)		R2 fast trip (s)		DSDR margin to last fast trip (ms)
	Conventional	DSDR	Conventional	DSDR reverse	
0 %	+4	+205	no trip	no trip	+205
10 %	−5	+175	no trip	no trip	+175
25 %	−17	+139	no trip	0.518	−279
37 %	−24	+116	1.109	0.231	−14
50 %	−31	+96	0.551	0.140	+57
75 %	−42	+65	0.250	0.080	+65
100 %	−51	+43	0.160	0.060	+43

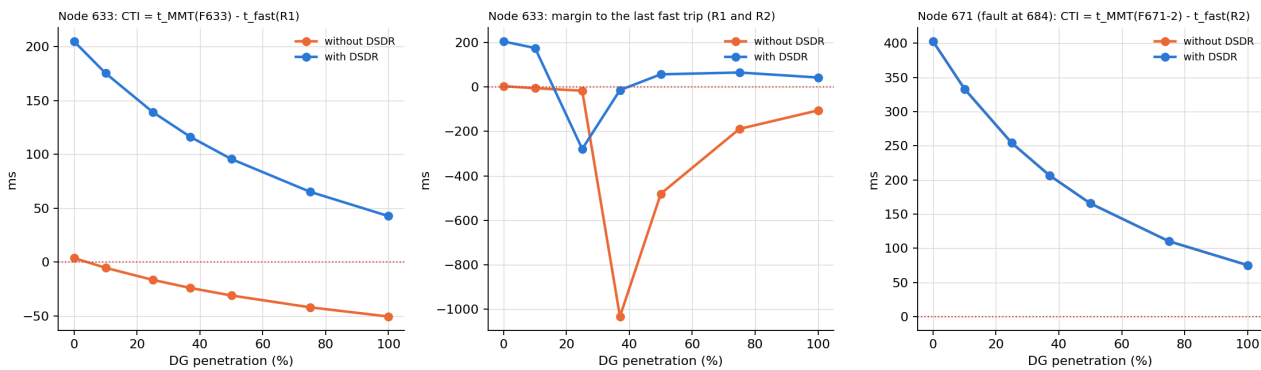
- **CTI against R1 stays positive at every level** with the DSDR (+205 → +43 ms) instead of turning negative (+4 → -51 ms). The DG still erodes it (by 162 ms), but from a much larger start.
- **Margin to the last fast trip** (fuse melting time minus the later of R1's and R2's fast trips — the strict test, because the DG current keeps heating the fuse until R2 opens): positive at 0–10 % (the DG current is too small to matter) and from 50 % up. At 0–10 % the DG current through F633 (66–161 A) is below the fuse's melting range, so the fuse survives even though R2 does not trip.
- **The weak spot at 25–37 %.** The reverse group was designed for 100 % DG. At low penetration the DG current (416–610 A) is only just above its 300 A start, so the reverse fast trip is slow (0.518 and 0.231 s) and the margin is negative (−279 and −14 ms). The conventional R2 is worse there: it does not trip at 25 % and trips only after 1.109 s at 37 % (−1033 ms). The remedy is adaptive: a reverse group selected for the actual DG output — listed as future work.



TCC of the fault at 633 with the DSDR protection. Left: 100 % DG. Right: 25 % DG. Grey: the old F633 250E, for comparison.

How to read it — the two actions of the DSDR are both visible:

- **Fuse revision = the green curve moved up and right** (grey 250E → green 400E). At the same fuse current the melting time is longer: at 100 % DG 0.149 s instead of 0.055 s. The fuse square is now *above* the R1 circle again: CTI against R1 = +43 ms.
- **Reverse group = the solid orange curve.** R2 carries the DG share (1500 A in reverse at 100 %). On the reverse curve it trips in 0.060 s; on the conventional forward curve (dashed orange) it would take 0.160 s, longer than the fuse's 0.149 s. So R1 removes the grid share at 0.106 s and R2 the DG share at 0.060 s — both before the fuse melts: the margin to the last fast trip is +43 ms.
- **25 % DG (right):** the fuse and R1 are still fine (+140 ms), but the DG share is only 416 A, just above the start of the reverse curve (300 A), where the curve is steep and slow: R2 trips at about 0.52 s, after the fuse's 0.24 s. This is the weak spot of a reverse group designed for 100 % DG; the conventional R2 does not trip at all.



Left: CTI at 633 against R1. Middle: margin to the last fast trip at 633 (R1 and R2). Right: CTI for the fault at 684 — identical for both schemes.

4.3 External fault at 684: the DSDR does not help

The CTI at 671 is identical in both schemes (+403 → +75 ms). For a fault downstream of R2 the current through R2 is forward, so the forward group is used, which is the same in both schemes, and F671-2 is not revised. The DG at 692 feeds this fault directly, with no recloser in its path, so no setting of R2 can remove the DG share. This is a limit of the DSDR concept itself; it is also why the time-sequence check shows “DG keeps feeding” for all 21 cells below R2.

4.4 The same effect in the full coordination study (DG 4.05 MVA)

Protection	Cells held (of 39)
No DG, design fuses	39
DG, conventional R2, design fuses	24
DG, DSDR, design fuses	25
DG, conventional R2, revised fuses	31
DG, DSDR, revised fuses	38

Example (three-phase fault near 632, R2 carries 1777 A in reverse): single setting, R2 fast 0.121 s and F633 melts at 0.100 s — lost; dual setting, R2 reverse fast 0.052 s — held.

In one sentence: The DSDR gives R2 a reverse group that cuts the DG's share of the fault current quickly, and the fuse revision restores the fuse's head-start over R1; together they keep the CTI positive for faults between the grid and the DG, but they cannot help faults downstream of R2, where the DG feeds the fault directly.

5. Likely questions

Why does the CTI decrease and not increase? Because the fuse sees grid + DG and the recloser only grid; the fuse becomes relatively faster. If the recloser carried the larger current (fault upstream of the fuse's source side), the opposite would happen, but in this topology the fuse is always the device nearer the fault.

Does the DSDR stop the CTI from falling? No. It keeps it positive. The DG still erodes the margin (633: –162 ms from 0 to 100 %), but the DSDR and the fuse revision start from a much larger margin and the reverse group removes the DG current.

Why is the DSDR worse than conventional at 25 % in the margin column? It is not worse in practice: the conventional R2 never trips there, so the margin column ignores it, while the slow reverse trip of the DSDR (0.518 s) is counted. The column is a strict, conservative measure. The real cure is a reverse group adapted to the DG output.

What would you do for faults below R2? The DG must be disconnected or a recloser/protection placed at the DG connection, or the fuses below R2 sized for grid + DG current; the DSDR alone cannot solve it.